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OVERCOMING THE UNCERTAINTY CHALLENGES IN FLOOD RAPID MAPPING WITH SAR DATA

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ABSTRACT

The escalating intensity and frequency of floods, exacerbated by global climate change, emphasize the urgent need to address the growing risks of floods. Rapid and precise flood detection is paramount for efficiently responding to emergencies and executing disaster relief measures, enabling swift reactions to flood disasters and minimizing the losses incurred thus. The 2024 IEEE GRSS Data Fusion Contest Track 1 is centered on leveraging multi-source remote sensing data, particularly synthetic aperture radar (SAR) data, to classify flood and non-flood areas. In this contest, we acknowledge the significance of managing uncertain predictions and present an efficient Uncertainty-Aware Fusion Network (UAFNet). Specifically, we build on the traditional encoder-decoder architecture, initially employing the pyramid visual transformer (PVT) as a feature extractor. Subsequently, we apply a typical decoding strategy, namely the feature pyramid network, to obtain a flood extraction map with relatively high uncertainty. Furthermore, leveraging the uncertain extraction map, we introduce an Uncertainty Rank Algorithm to quantify the uncertainty level of each pixel of the foreground and background. We seamlessly integrate this algorithm with our proposed Uncertainty-Aware Fusion Module, enabling level-by-level feature refinement and ultimately yielding a refined extraction map with minimal uncertainty. Employing the proposed UAFNet, we utilize diverse versions of PVT as encoders to train multiple UAFNets. Additionally, we enhance our approach with online testing augmentation and multi-model fusion strategy, aiming to enhance the final flood extraction accuracy. Our technical solution has exhibited outstanding performance, earning the first-place ranking in the 2024 IEEE GRSS Data Fusion Contest Track 1 and achieving an impressive F1 score of 82.985% on the official test set.

***Index Terms*—** SAR, flood mapping, semantic segmentation, uncertainty-based theory

1. INTRODUCTION

Flood disasters stand as one of the most prevalent natural calamities in the contemporary era. Their widespread impact and devastating losses frequently pose a grave threat to national economies, along with the safety of lives and properties. Hence, flood detection research has emerged as a pivotal topic in natural disaster studies. With the remarkable progress [1, 2, 3] in remote sensing (RS) technology, RS images have become a crucial source of data for acquiring flood information. Their swiftness, strong timeliness, and extensive repeat observations render them indispensable in this domain.

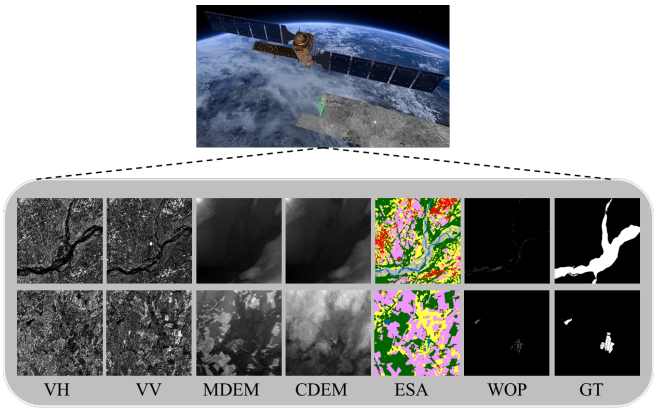


Fig. 1: Illustration of multi-source remote sensing images used in the DFC Track1.

2024 IEEE GRSS Data Fusion Contest Track 1 aims to develop data fusion algorithms that generate flood maps by processing spatial data from a variety of sources. As shown in Fig. 1, in this contest, Synthetic Aperture Radar (SAR) images (including VH and VV), digital elevation models (including Merit DEM and Copernicus DEM), cover map (ESA World Cover Map [4]), and water occurrence probability (WOP) are provided to classify flood and non-flood areas.

Given the vast array of land features captured in RS images, the intricate nature of contextual information, and the inherent irregularities in SAR images, we have incorporated the concept of uncertainty [5] to tackle these challenges. In this paper, we introduce an efficient Uncertainty-Aware Fusion Network (UAFNet). This network builds upon the established Encoder-Decoder architecture [6], initially leveraging the pyramid visual transformer (PVT [7, 8]) as the feature

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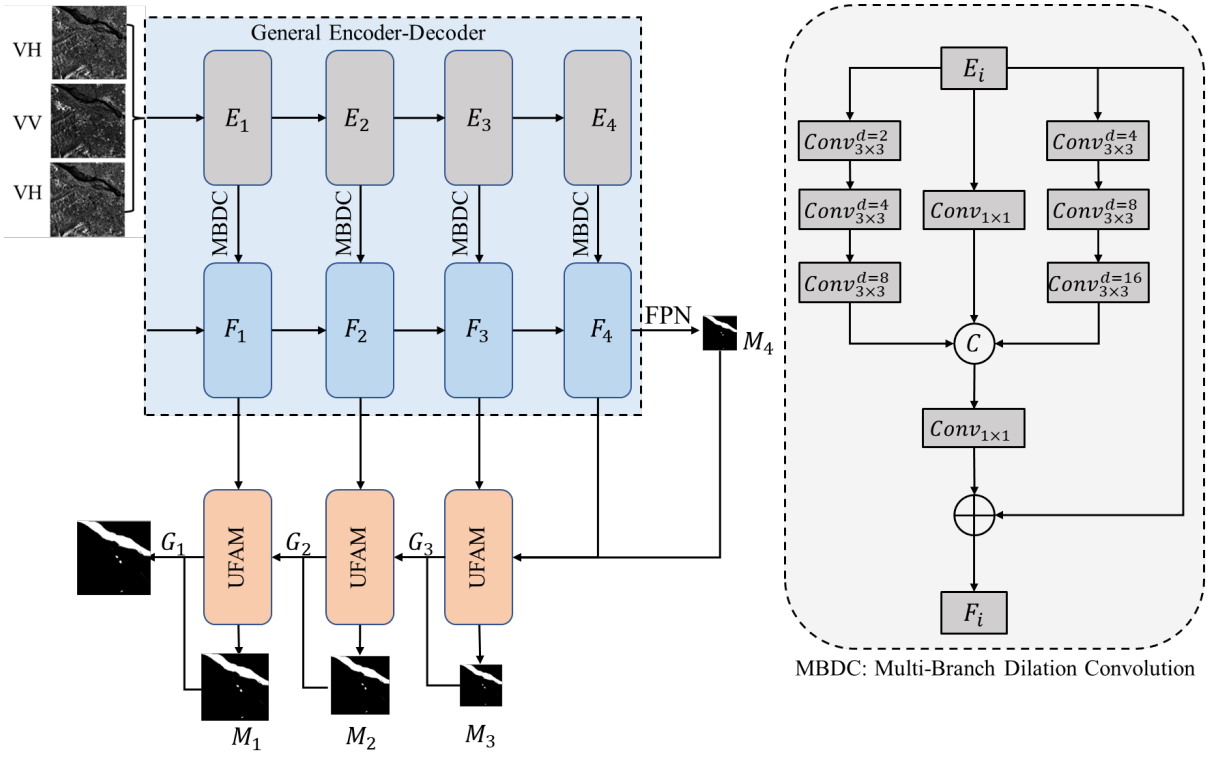


Fig. 2: The structure of the Uncertainty-Aware Fusion Network.

extractor. Subsequently, we employ a typical decoding strategy, specifically the feature pyramid network (FPN [9]), to generate a flood extraction map that incorporates a relatively high level of uncertainty. To further refine this extraction map, we introduce an Uncertainty Rank Algorithm that quantifies the uncertainty level of each pixel, distinguishing between the foreground and background. This algorithm is seamlessly integrated into our proposed Uncertainty-Aware Fusion Module, enabling gradual feature refinement and ultimately delivering a refined extraction map with minimal uncertainty. Using various versions of PVT as encoders, we train multiple instances of UAFNet. Additionally, we enhance our approach through online testing augmentation and a multi-model fusion strategy, aiming to boost the accuracy of flood detection. Our technical solution has proven to be highly effective, achieving a commendable F1 score of 82.985% on the official test set.

2. METHODOLOGY

2.1. Feature Extraction

Aiming to eliminate the uncertainty of the final extraction map as much as possible, we propose the Uncertainty-Aware Fusion Network (UAFNet). As shown in Fig.2, we first adopt a general encoder-decoder network to get a relatively uncertain extraction map. Regarding the general encoder-decoder part, we adopt PVT-V2 [8] as the encoder backbone to extract multi-level features from the input image and introduce a multi-branch dilation convolution blocks (MBDC) to enhance the encoded features ($E_i, \{i = 1, 2, 3, 4\}$), and use a typical cross-fusion strategy (Feature Pyramid Network (FPN [9])) to obtain a relatively uncertain extraction map M_4, the whole process can be represented as:

$$\begin{aligned} F_i &= MBDC(E_i), i = 1, 2, 3, 4, \\ M_4 &= FPN(F_1, F_2, F_3, F_4). \end{aligned} \tag{1}$$

Based on the output features ($F_i, \{i = 1, 2, 3, 4\}$) and uncertain extraction map M_4, our UAFNet acts as a decoder strategy to deal with the general flood extraction challenges and output a refined extraction map with low uncertainty.

2.2. Uncertainty-Aware Fusion Module

The uncertainty caused by the intricate backgrounds and various scales remains. Therefore, we present the uncertainty-aware fusion module (UAFM) introduced in UANet [10] to tackle the high uncertainty issue.

Specifically, As we all know, all the deep learning approaches output the extraction results by using the $Softmax$ or $Sigmoid$ function to allocate the corresponding probability for each pixel, which can be directly used to reflect the uncertainty of the model in its predictions. As mentioned before, in RS images, some floods are not salient enough and do not appear frequently, which will result in the uncertainty of model. To overcome such a uncertainty problem, we directly use the $Sigmoid$ function to get the corresponding probabilities of all pixels in the extraction map M from spatial per-

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spective, then we subtract all values of the probability map with 0.5 to measure the uncertainty belonging to foreground flood (U_f) and meanwhile subtract 0.5 with all values of the probability map to measure the uncertainty belonging to background (U_b),

$$\begin{aligned} U_f &= Sigmoid(M) - 0.5, \\ U_b &= 0.5 - Sigmoid(M). \end{aligned} \tag{2}$$

Subsequently, we rank the uncertainty of the foreground and background into five levels using the Uncertainty Rank Algorithm (URA), that is, the range of $[-0.5, 0)$ represents not in consideration (rank 0), the range of $[0, 0.1)$ indicates the highest uncertainty (rank 5), the range of $[0.1, 0.2)$ represents the relatively high uncertainty (rank 4), the range of $[0.2, 0.3)$ represents the central uncertainty (rank 3), the range of $[0.3, 0.4)$ indicates moderately low uncertainty (rank 2), and the range of $[0.4, 0.5]$ denotes the lowest uncertainty (rank 1). We then assign the corresponding uncertainty levels as weights to the pixels, with the principle of attaching higher weights to pixels with higher uncertainty, so as to pay more attention to uncertain areas. We denote URA as:

$$\mho(i,j) = \begin{cases} \lfloor \frac{0.5 - U_{i,j}}{0.1} \rfloor, U_{i,j} >= 0, \\ 0, U_{i,j} < 0, \end{cases} \tag{3}$$

where $U_{i,j}$ means the pixel in i_{th} row and j_{th} column of U_f or U_b.

Therefore, after using URA to allocate the uncertainty level about the uncertainty maps of the foreground and the background, respectively, we can obtain the foreground uncertainty rank map (R_f) and the background uncertainty rank map (R_b).

$$\begin{aligned} R_f &= URA(Sigmoid(M) - 0.5), \\ R_b &= URA(0.5 - Sigmoid(M)), \end{aligned} \tag{4}$$

We take the fusion of the highest F_4 and F_3 for example to illustrate the whole fusion process. Specifically, the inputs of UAFM in this layer are the feature F_4, the F_3, and the uncertain extraction map M_4. Regarding the uncertainty-aware enhancement, on the one hand, we apply URA to M_4 so that we can get the corresponding foreground uncertainty rank map (R_f^4) and background uncertainty rank map (R_b^4). Then we directly use F_4 to multiply with them to highlight the uncertain pixels from both the foreground and background perspectives. Subsequently, we concatenate these two enhanced features and recover its original channel to get F_4^u by a 1×1 convolution operation.

$$F_4^u = Conv_{1\times1}(Concat(R_f^4 * F_4, R_b^4 * G_4)), \tag{5}$$

On the other hand, we use the nearest neighbor interpolation method to upsample R_f^4 and R_b^4 to the same size as F_3, and use the same operation to highlight F_3 as the enhancement of F_4, and we can get F_3^u.

$$F_3^u = Conv_{1\times1}(Concat(Up(R_f^4) * F_3, Up(R_b^4) * F_3)), \tag{6}$$

Finally, we upsample F_4^u to match the size of F_3^u, concatenate them together, and use a 3×3 convolution operation to get the fused feature G_3, which can output the less uncertain extraction map M_3 by a 3×3 convolution operation.

$$\begin{aligned} G_3 &= Conv_{3\times3}(Concat(F_3^u, Up(F_4^u))), \\ M_3 &= Conv_{3\times3}(G_3), \end{aligned} \tag{7}$$

We employ the uncertainty-aware fusion module (UAFM) to fuse the features G_i and F_{i-1} layer-by-layer and decode the fused feature to output the corresponding certainty-improved map M_{i-1}.

With such a UAFM, we can utilize M_3 to fuse G_3 and F_2 and achieve output G_2 and M_2, and utilize M_2 to fuse G_2 and F_1 and output G_1 and M_1, where M_1 can be viewed as the final refined extraction map with the lowest uncertainty.

On the whole, we use the simple binary cross-entropy (BCE) loss function to supervise all the outputs, and the overall loss is:

$$Loss = \sum_{i=1}^{4} BCE(M_i, GT), \tag{8}$$

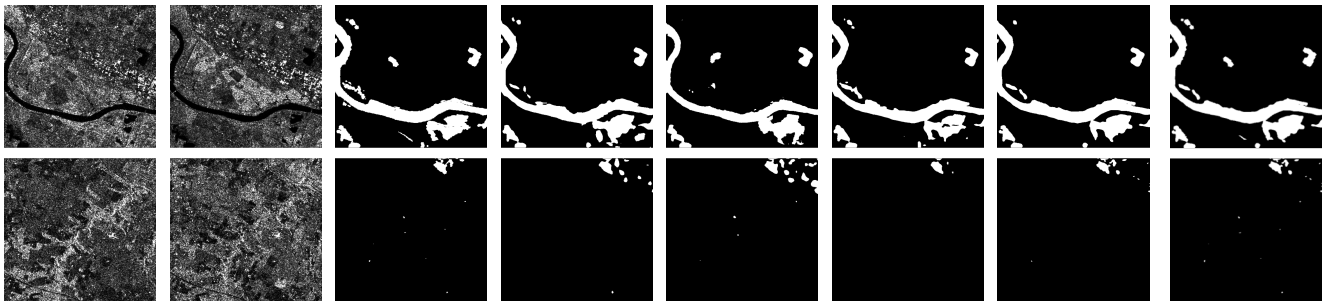
where GT represents the ground truth.

3. EXPERIMENTS

3.1. Experimental Settings

For a comprehensive evaluation of our proposed model, all experiments were conducted using PyTorch 1.8.1 (CUDA 11.1) on an NVIDIA GeForce RTX 3090 GPU with 24GB of memory. During the training phase, we employed the AdamW [14] optimizer and applied the cosine annealing strategy to adjust the learning rate, starting from an initial learning rate of 10^{-4}. To create a three-dimensional input for leveraging the PVT-v2 pre-trained on ImageNet [15] and enhancing the feature representation capability, we concatenated VH, VV, and VH together. Furthermore, by utilizing various versions of PVT as encoders (PVT-V2-B2, PVT-V2-B3, PVT-V2-B4, PVT-V2-B5), we trained four instances of UAFNet and adopted the multi-model fusion strategy to improve the extraction performance. Additionally, upon analyzing the official training data, we discovered the crucial role of provided land cover data in post-processing. To enhance the post-processing process, we merged the predicted results of our strategy on the test set with the designated water regions from the land cover data. Finally, it is worth noting that we applied twelve enhancements to each image in the training set by using six rotation angles and six flip modes, effectively expanding the dataset.

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(a) VH (b) VV (c) GT (d) HRNet [11] (e) UNet [12] (f) SF [13] (g) UAFNet (h) Ours

Fig. 3: Visual Comparison between UAFNet and the compared methods. SF is short for SegFormer, and Ours indicates the visual results of **Multi-UAFNet+post-processing.**

Table 1: Quantitative results on the official test dataset. * denotes the absence of UAFM.

Method	Test (%) $F1 \uparrow$
UNet (ResNet-50)	73.935
SegFormer (MIT-B5)	74.996
UNet (PVT-V2-B2)	75.363
UAFNet* (PVT-V2-B2)	76.808
UAFNet (PVT-V2-B2)	78.756
UAFNet (PVT-V2-B3)	78.773
UAFNet (PVT-V2-B4)	79.069
UAFNet (PVT-V2-B5)	80.824
Multi-UAFNet	<u>81.890</u>
Multi-UAFNet + Post-Processing	**82.985**

3.2. Visual Comparison

As depicted in Fig. 3, we selected two typical examples to illustrate the effectiveness of our proposed UAFNet. It can be seen that compared to other methods, there are some slight issues in extracting details. Our method can achieve results that are closer to the true labels in terms of details. Due to the limitation of resolution, we can also see from VH and VV images that it is difficult to accurately determine whether certain pixel areas belong to flood. However, to some extent, our uncertainty strategy can overcome this challenge.

3.3. Quantitative Comparison

As illustrated in Tab. 1, it is evident that the different methods and backbone network versions exhibit varying levels of performance in the flood extraction task. The traditional ResNet-50-based [16] UNet [12] method achieves an F1 score of 73.935%, while SegFormer [13] slightly outperforms it with a score of 74.996%. However, when combining UNet with the PVT-V2-b2 backbone network, there is a noticeable improvement in performance, with an F1 score of 75.363%. This indicates that the PVT series of backbone networks have a positive impact on flood extraction tasks. At the same time, Compared to UNet (PVT-V2-B2), UAFNet (PVT-V2-B2) demonstrates a significant improvement in F1 score, achieving 78.756%. Meanwhile, we can observe that the improvement brought by UAFM can directly enhance the model's performance from an F1 score of 76.808% to 78.756%, fully demonstrating the effectiveness of UAFM. As we upgrade the backbone network version from PVT-V2-B2 to PVT-V2-B5, the performance of UAFNet gradually improves, reaching a peak of 80.824%. This underscores the superiority of the UAFNet approach in handling flood extraction tasks and its scalability with more capable backbone networks. Furthermore, we explore a multi-model fusion strategy, known as Multi-UAFNet, which combines the results from four models ranging from PVT-V2-B2 to PVT-V2-B5. This strategy further enhances the accuracy of flood extraction, achieving an F1 score of 81.890%, surpassing the best performance of a single model. Finally, by introducing a post-processing strategy, we optimize the results of Multi-UAFNet, achieving the highest F1 score of 82.985%. This not only demonstrates the effectiveness of the UAFNet approach but also highlights the importance of post-processing in enhancing the precision of flood extraction.

In summary, our UAFNet method, coupled with the PVT series of backbone networks, significantly improves the performance of flood extraction tasks. Through multi-model fusion and post-processing, we further enhance the accuracy of flood detection, providing robust technical support for flood monitoring and early warning systems.

4. CONCLUSION

In this paper, we have proposed an Uncertainty-aware Fusion Network (UAFNet) to address the uncertainty issues in flood mapping from SAR data. The experimental results demonstrated the performance of our UAFNet, achieving first place in the 2024 IEEE GRSS Data Fusion Contest Track 1. In the future, we will continue to explore the approach to achieve high-precision flood extraction on a global scale.

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